

# TITLE

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## Abstract

ABSTRACT

**Index Terms:** KEYWORDS

## 1. Models

The detector suite is chosen to probe distinct failure modes rather than to maximize benchmark score alone. It targets contamination-prior sensitivity, dimensionality stress and distance concentration, local versus global anomaly structure, and dependence on marginal-independence assumptions. The resulting mix covers boundary learning, random isolation, local density ratio estimation, density-connectivity clustering, and marginal distribution scoring, which makes it possible to attribute performance changes to the underlying mechanism of each detector under a shared preprocessing and evaluation protocol.

**OCSVM** OCSVM is the boundary-based baseline. In its RBF (Radial Basis Function) form, it implicitly maps the normal training data into a high-dimensional space and attempts to draw a tight, flexible "envelope" or boundary around it, separating the normal data from the origin. The hyperparameter  $\nu$  controls the strictness of this envelope by dictating the maximum allowed fraction of training points that can fall outside of it (acting as errors or anomalies). Intuitively, if a new point falls outside this learned boundary, it is flagged as an anomaly [8].

**IForest** IForest represents partition-based detection. It works on the premise that anomalies are "few and different." The algorithm builds a forest of random decision trees by randomly selecting a feature and a random split value between the maximum and minimum values of that feature. Because anomalies are sparse and far from normal data clusters, it takes very few random splits to isolate an anomaly into its own leaf node. Conversely, normal points are densely packed and require many more splits to be isolated. The shorter the average path length from the root to the leaf, the more anomalous the point [6].

**LOF** LOF captures local anomaly structure by comparing a point's local density with that of its neighbors. Unlike global methods that look at the entire dataset at once, LOF recognizes that datasets often have clusters of varying densities. It calculates the density around a specific point and compares it to the average density of its  $k$ -nearest neighbors. If a point has a significantly lower density than its neighbors, it means it is relatively isolated *for its specific neighborhood*, even if it isn't the furthest point globally. This makes LOF highly sensitive to subtle, local deviations [1].

**DBSCAN** DBSCAN uses density connectivity to separate

clusters from noise. It groups together points that are closely packed together (i.e., having a minimum number of neighbors within a specific radius,  $\epsilon$ ). Points that sit in low-density regions and cannot be reached from any dense cluster are simply left out in the cold. Instead of assigning an anomaly score on a sliding scale, DBSCAN provides a stark binary contrast: a point is either part of a dense, connected component (normal) or it is labeled as unclustered "noise" (an outlier) [2].

**ECOD** ECOD is the marginal-distribution baseline. It looks at the dataset one feature at a time, calculating the Empirical Cumulative Distribution Function (CDF) for each. Intuitively, it asks: "What is the probability of seeing a value this extreme, or more extreme, in this specific feature?" It then aggregates these probabilities (the "tail evidence") across all dimensions for a given point. Because it looks at features independently and relies on empirical probabilities rather than assuming the data fits a specific shape (like a bell curve), it serves as a highly efficient, parameter-light reference model [5].

**HBOS** HBOS also relies on marginal (feature-wise) evidence, but instead of using the continuous CDF like ECOD, it builds discrete histograms for each feature. To score a point, HBOS checks which histogram bin the point's value falls into for every feature. If a point frequently falls into very short bins (meaning very few other data points share that value), it accumulates a high anomaly score. Because it relies on histograms, it is exceptionally fast, but its accuracy is highly sensitive to how you define the bins (e.g., bin width and the number of bins) [4].

Table 1 lists the hyperparameter axes and candidate values for each detector.<sup>1</sup>

## 2. Datasets

### 2.1. Dataset Sources and Composition

The benchmark corpus combines tabular anomaly detection datasets from the ODDS library [7] with selected two-dimensional datasets from the Gagatewski clustering suite [3]. All datasets are standardized into a common schema with feature matrix  $X \in \mathbb{R}^{n \times d}$  and binary label vector  $y \in \{0, 1\}^n$ , where 1 denotes an anomalous observation. The corpus contains 37 datasets, spanning 1,086,545 total observations, with sizes ranging from  $n = 120$  to  $n = 567,498$ , dimensionality

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<sup>1</sup> $n_{\text{runs}}$  is the Cartesian size when unconditional axes are combined ( $\prod$  list lengths). One-class SVM mixes kernel-dependent parameters, so only valid library configurations are materialised rather than a naive full product. Section 3 explains which parts of this space enter the primary benchmark and which enter the focused studies.

Table 1: Hyperparameter axes by model;  $n\_runs$  is the Cartesian axis count.

| Model   | Parameter                                 | Values                                                                 | Motivation                                                                                                                                  | $n\_runs$ |
|---------|-------------------------------------------|------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| OCSVM   | kernel                                    | {rbf, poly, sigmoid, linear}                                           | Compare boundary families and kernel scalability.                                                                                           | †         |
|         | nu                                        | {0.01, 0.05, 0.10, 0.20, 0.35, 0.50}                                   | Upper bound training errors / fraction of outliers.                                                                                         |           |
|         | gamma                                     | {scale, auto, 0.001, 0.01, 0.1, 1.0, 10.0}                             | RBF / poly / sigmoid scaling; diagnoses kernel collapse regimes.                                                                            |           |
|         | degree<br>coef0                           | {2, 3, 4}<br>{0.0, 1.0}                                                | Polynomial depth (poly kernel).<br>Affine term for poly / sigmoid kernels.                                                                  |           |
| IForest | n_estimators<br>max_samples               | {50, 100, 200, 500}<br>{auto, 32, 64, 128, 256, 512}                   | Check ensemble convergence and variance reduction.<br>Probe $\psi$ / subsampling depth per Liu et al.; auto clamps to $\min(n, 256)$ .      | 1008      |
|         | contamination                             | {0.01, 0.05, 0.10, 0.20, 0.35, 0.50}                                   | Prior-dependent threshold shaping.                                                                                                          |           |
|         | max_features<br>bootstrap<br>random_state | {0.5, 0.75, 1.0}<br>{True, False}<br>{42}                              | Feature bagging breadth for each tree split.<br>Compare with vs. without replacement.<br>Locked reproducibility knob in exploratory sweeps. |           |
|         |                                           |                                                                        |                                                                                                                                             |           |
| LOF     | n_neighbors                               | {5, 10, 15, 20, 30, 50, 75, 100}                                       | Locality scale controlling reachability densities.                                                                                          | 480       |
|         | metric                                    | {euclidean, manhattan, minkowski, cosine}                              | Geometry stress for curse-of-dimensionality probes.                                                                                         |           |
|         | p<br>contamination                        | {1, 2, 3}<br>{0.01, 0.05, 0.10, 0.20, 0.35}                            | Minkowski exponent when metric=minkowski.<br>Prior-dependent thresholds on negative outlier factors.                                        |           |
|         | novelty                                   | {False}                                                                | Transductive training scores (no withheld calibration split).                                                                               |           |
| DBSCAN  | eps                                       | $eps_{knee} \times \{0.5, \dots, 2.0\}$ , six multipliers total        | Knee anchored on sorted $k$ -distance; multipliers perturb neighborhood radius.                                                             | 72        |
|         | min_samples<br>metric                     | {3, 5, 10, 15, 20, 30}<br>{euclidean, manhattan}                       | Core-point cardinality control.<br>Metric-dependent density connectivity tests.                                                             |           |
|         | algorithm                                 | {auto}                                                                 | sklearn NN backend fallback (BallTree/KD/BF).                                                                                               |           |
| ECOD    | contamination                             | {0.01, 0.05, 0.10, 0.20, 0.35, 0.50}                                   | Tail-threshold sliders (scores unaffected).                                                                                                 | 6         |
| HBOS    | n_bins                                    | {5, 10, 20, 30, 50, auto}, $auto \rightarrow \lfloor \sqrt{n} \rfloor$ | Histogram fidelity vs. sparsity; PyOD heuristic for auto.                                                                                   | 240       |
|         | alpha                                     | {0.0, 0.1, 0.2, 0.5}                                                   | Laplace smoothing of empty bins ( $\alpha \in [0, 1)$ constraint).                                                                          |           |
|         | tol<br>contamination                      | {0.1, 0.5}<br>{0.01, 0.05, 0.10, 0.20, 0.35}                           | Numerical tolerance for tails when mapping to bins.<br>Prior thresholds on aggregated log-scores.                                           |           |

†OCSVM: the RBF core alone has  $6 \nu \times 7 \gamma = 42$  combinations; additional kernels extend the table. Focused one-class SVM analyses (Sobol sampling, Latin Hypercube exploration, and the ionosphere oracle sweep) are summarised in Section 3.

from  $d = 2$  to  $d = 400$ , and natural contamination rates from 0.0% to  $\approx 35.9\%$  (canonical label incidence in Appendix A).

The ODDS subset comprises 26 multidimensional tabular datasets, most available via the ADBench mirror [7]. The 2D subset comprises 11 datasets from Gagolewski et al. (v1.1.0), selected to expose solver sensitivity to geometric structure (concentric rings, interleaved curves, sparse neighborhoods, and boundary ambiguity).

## 2.2. Corpus Design Rationale

The corpus is intentionally heterogeneous to expose distinct detector failure modes:

**High-dimensional stress** `arrhythmia` ( $d=274$ ), `musk` ( $d=166$ ), `speech` ( $d=400$ ), and `mnist` ( $d=100$ ) test kernel collapse and distance-metric degradation.

**Extreme scale** `cover` ( $n=286,048$ ), `http` ( $n=567,498$ ), `smtp` ( $n=95,156$ ), and `shuttle` ( $n=49,097$ ) test algorithmic scalability and numerical stability.

**Prior mismatch** `breastw` (35.0%), `ionosphere` (35.9%), `pima` (34.9%), and `satellite` (31.6%) test methods sensitive to contamination assumptions when the prior is far from typical (1–5%).

**Geometric complexity** Gagolewski 2D datasets expose solver sensitivity to local versus global anomaly structure, symmetry, and boundary ambiguity.

## 2.3. Data Acquisition and Preprocessing

Dataset acquisition and preprocessing follow a uniform reproducible protocol:

**Download and validation** Raw artifacts are downloaded from verified public sources (ADBench mirrors for ODDS; official GitHub release for Gagolewski v1.1.0). Each dataset is validated against expected row count, feature count, and contamination rate.

**Canonicalization** All datasets are converted to a common tabular format (CSV or NumPy) with explicit binary labels. Missing values are documented and, for `arrhythmia`, imputed using an iterative procedure (see below).

**Default preprocessing** All datasets (except those with dedicated pipelines) are preprocessed with robust scaling (median center, IQR normalize) applied only to numeric features. This preserves low-d interpretability while reducing outlier influence on scale estimates.

**Arrhythmia-specific preprocessing** Archival `arrhythmia` documentation cites  $\sim 0.33\%$  incomplete entries (nominals mixed with continuous ECG signals); the ODDS-style mirror canonicalised here does not leave those artefacts as CSV-level pandas NaNs, hence Appendix A reports a 0 missing fraction. Preprocessing branches still execute iterative Bayesian Ridge imputation on any latent gaps surfaced during engineering. Steps are (1) imputation plus encoding pipeline (10 imputer iterations where needed), (2) one-hot encoding, and (3) a high-dimensional encoded view alongside PCA-95. In the main benchmark, LOF and DBSCAN use the PCA-95 view only, while IForest, OCSVM, ECOD, and HBOS also see the engineered path (paired tasks also add PCA-95 for everyone).

## 2.4. Algorithm-Specific Subsampling

Computational cost varies widely across algorithms. To ensure tractability while maintaining methodological integrity, the following protocol is applied:

**No subsampling** IForest, ECOD, and HBOS run on full dataset due to their favorable time complexity ( $O(n \log n)$  or  $O(nd \log n)$ ). Running on full data for these methods is itself a contribution, demonstrating industrial-grade scalability.

**Stratified subsampling** OCSVM, LOF, and DBSCAN (all quadratic or worse) are capped at  $n=20,000$  observations when the full dataset exceeds this size. Stratified random sampling preserves the contamination ratio. This cap applies to four datasets: `cover`, `http`, `smtp`, and `shuttle`.

Crucially, anomaly labels are reserved for evaluation only and never enter the fitting or hyperparameter-selection pipeline. This ensures a faithful unsupervised protocol. Detailed corpus-level and per-dataset descriptive statistics are reported in Appendix A.

# 3. Experiments

This section states the experimental plan in plain terms. The hyperparameter grid in Table 1 defines what we *could* vary; the runs below describe what we *actually* run, and why.

## 3.1. Two layers: benchmark and deep dives

The study has two layers. The **primary benchmark** fits each detector once per dataset with a single unsupervised default (no labels for model selection). Defaults are chosen to match common library prescriptions: Isolation Forest with a medium-sized ensemble and fixed subsample cap, LOF with a moderate neighbour count, DBSCAN with a knee-estimated neighbourhood radius, RBF one-class SVM with a moderate  $\nu$ , and library defaults for ECOD and HBOS. This layer answers: “how do these methods behave off the shelf?” Stratified subsampling for the slower detectors, when  $n$  exceeds the cap, is fixed in Section 2.

The **focused studies** widen selected knobs only where the corpus is likely to reveal something about robustness—geometry for local methods, neighbourhoods for clustering, contamination priors for tail models—and repeat each setting with five random seeds  $\{0, 1, 2, 3, 42\}$  so reported spread reflects optimisation noise, not one-off luck. These studies do *not* brute-force every cell of Table 1 on every benchmark; that table is intentionally large so that Section 4 and the appendix can cite consistent axis definitions.

## 3.2. Focused hyperparameter sweeps

**Isolation Forest** on `breastw`: forest sizes  $\{50, 100, 200, 500\}$  trees, subsample sizes 128 and 256, contamination priors 0.05 and 0.10 for thresholding, bagging bootstrap off, five seeds each.

**Local outlier factor** on `satellite`: eight neighbour counts spaced from 5 to 100, Euclidean distance, five seeds; the same neighbour grid is repeated under Manhattan, cosine, and Minkowski ( $p=3$ ) metrics with identical seeds.

**DBSCAN** on `satellite`: six scalings of a knee-derived radius, six `min_samples` values from sparse to denser cores, Euclidean or Manhattan geometry, five seeds.

**ECOD** on `pima`: six contamination assumptions for translating scores into decisions.

**Histogram-based detection** on `satellite`: five discrete bin layouts (the automatic bin-count heuristic is switched off here), three Laplace-smoothing levels for empty bins, five seeds.

### 3.3. Sensitivity to hyperparameters (variance-based)

We estimate first-order Sobol indices for PR-AUC using a Saltelli-style quasi-random scheme with  $N_{\text{saltelli}} = 28$  base draws per dataset–model pair. Isolation Forest is analysed on `ionosphere`, `breastw`, and `smtp`; RBF one-class SVM on `breastw`, `ionosphere`, and `pima`. In parallel, eighty Latin Hypercube draws explore  $\nu$  and  $\gamma$  pairs from the categorical RBF core on the same three SVM probe datasets—a complementary view that does not require treating  $\gamma$  as a continuous surrogate.

### 3.4. Oracle sweep and bootstrap stability

As a retrospective upper bound on RBF one-class SVM tuning, every  $\nu \times \gamma$  pair from the categorical RBF grid is evaluated on `ionosphere` with five seeds (210 pooled fits total).

Isolation Forest stability on `thyroid`: forty-eight hyperparameter tuples are picked deterministically from a dense Cartesian surrogate; each tuple receives ten stratified bootstrap replicates that retain roughly 90% of points. We favour tuples with high mean PR-AUC among replicates whose bootstrap standard deviation  $\sigma_{\text{boot}}$  stays at or below 0.03; if none qualify, we keep the strongest mean-score candidate regardless.

## 4. Evaluation Methodology

The evaluation protocol separates pointwise performance reporting from robustness analysis. Metrics summarize classification quality and ranking quality under a fixed contamination threshold, while sensitivity analysis measures how much performance changes under hyperparameter variation and resampling.

### 4.1. Metrics

All metrics are computed at the natural contamination threshold, so each method labels exactly the expected fraction of points as outliers. This keeps threshold selection fixed across datasets and makes the comparison directly about score quality rather than post-processing.

**Accuracy** retained as a baseline summary, but interpreted cautiously because it can be dominated by the majority class under extreme imbalance.

**Precision** measures the fraction of predicted outliers that are truly anomalous, which is useful when false positives are costly.

**Recall** measures how many true outliers are recovered, which is important when missing anomalies is expensive.

**ROC-AUC** evaluates ranking quality independently of class prevalence, so it is useful for cross-dataset comparison.

**PR-AUC** emphasizes performance on the positive class and is therefore more informative than ROC-AUC under heavy imbalance.

**MCC** summarizes all four confusion-matrix cells in a single balanced score, which makes it robust under skewed labels.

**F1** combines precision and recall into a single thresholded score at the natural contamination cut.

### 4.2. Sensitivity and robustness reporting

Robustness summaries accompany the main metrics. They address (i) how much PR-AUC moves under reasonable hyperparameter changes, and (ii) how stable scores are under bootstrap resampling. All schedules are in Section 3; for reproducibility the primary defaults are Isolation Forest ensemble size 200 with `max_samples = 256` and `bootstrap off`, LOF with 20 neighbours, DBSCAN knee  $\epsilon$  with `min_samples = 5`, RBF one-class SVM with  $\nu = 0.1$ , and library defaults for ECOD and HBOS. Anomaly labels are used for evaluation only and never to pick hyperparameters.

## 5. References

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## A. Dataset Details

This section provides corpus-level and per-dataset descriptive statistics used in the experimental design and analysis. Each row reflects merged feature–label tables read straight from persisted canonical CSV files, matching the pipeline’s raw canonical slice (no robust scaling yet, no arrhythmia engineered matrix, no PCA exports). Missing fraction is total pandas NA cells divided by  $n_{\text{rows}} \times n_{\text{cols}}$  for that canonical table (features plus label)—Appendix percentages can disagree with older “% incomplete” aggregates: ODDS-derived `arrhythmia` emits no CSV-level NaNs and therefore shows 0.0000, while handbook figures still cite  $\sim 0.33\%$  masked cells historically.

### A.1. Corpus-Level Summary

Table 2 summarizes key corpus properties. The 37 datasets range from extremely small ( $n = 120$ ) to massive ( $n = 567,498$ ), and from low-d ( $d = 2$ ) to ultra-high-d ( $d = 400$ ), ensuring comprehensive coverage of failure modes.

### A.2. Per-Dataset Descriptive Statistics

Table 3 provides per-dataset statistics used to characterize solver stress and guide preprocessing decisions. Columns: `dataset_id` (name), `n_rows/n_features`

Table 2: *Corpus-level dataset summary. Statistics reflect raw datasets after canonicalization, before preprocessing.*

| Property                           | Value         |
|------------------------------------|---------------|
| Total datasets                     | 37            |
| ODDS datasets                      | 26            |
| 2D benchmark datasets (Gagolewski) | 11            |
| Total observations                 | 1,086,545     |
| Observation count range            | 120 – 567,498 |
| Dimensionality range               | 2 – 400       |
| Contamination rate range           | 0.0% – 35.9%  |

(size), contam. (anomaly rate), missing (fraction missing), outliers/inliers (counts), mean  $|\rho|$  (mean absolute correlation; 0=independent, 1=collinear), CR (distance concentration ratio;  $\gg 1$ =intact geometry,  $\lesssim 0.1$ =curse-of-dimensionality).

Table 3: *Per-dataset descriptive statistics.*

| dataset_id     | n_rows | n_features | contam. | missing | outliers | inliers | mean $ \rho $ | CR      |
|----------------|--------|------------|---------|---------|----------|---------|---------------|---------|
| annthyroid     | 7200   | 6          | 0.0742  | 0.0000  | 534      | 6666    | 0.2502        | 2.8144  |
| arrhythmia     | 452    | 274        | 0.1460  | 0.0000  | 66       | 386     | 0.0891        | 2.4601  |
| breastw        | 683    | 9          | 0.3499  | 0.0000  | 239      | 444     | 0.6019        | 2.2810  |
| cardio         | 1831   | 21         | 0.0961  | 0.0000  | 176      | 1655    | 0.2206        | 2.9517  |
| cover          | 286048 | 10         | 0.0096  | 0.0000  | 2747     | 283301  | 0.1932        | 2.5192  |
| glass          | 214    | 7          | 0.0421  | 0.0000  | 9        | 205     | 0.2825        | 2.8242  |
| http           | 567498 | 3          | 0.0039  | 0.0000  | 2211     | 565287  | 0.0558        | 4.2144  |
| ionosphere     | 351    | 32         | 0.3590  | 0.0000  | 126      | 225     | 0.2383        | 1.9646  |
| letter         | 1600   | 32         | 0.0625  | 0.0000  | 100      | 1500    | 0.2017        | 1.5442  |
| lympho         | 148    | 18         | 0.0405  | 0.0000  | 6        | 142     | 0.1681        | 2.4614  |
| mammography    | 11183  | 6          | 0.0232  | 0.0000  | 260      | 10923   | 0.2440        | 3.4408  |
| mnist          | 7603   | 100        | 0.0921  | 0.0000  | 700      | 6903    | 0.0761        | 0.9830  |
| musk           | 3062   | 166        | 0.0317  | 0.0000  | 97       | 2965    | 0.2931        | 1.3835  |
| optdigits      | 5216   | 64         | 0.0288  | 0.0000  | 150      | 5066    | 0.1210        | 1.0620  |
| pendigits      | 6870   | 16         | 0.0227  | 0.0000  | 156      | 6714    | 0.2629        | 1.3948  |
| pima           | 768    | 8          | 0.3490  | 0.0000  | 268      | 500     | 0.1717        | 5.7068  |
| satellite      | 6435   | 36         | 0.3164  | 0.0000  | 2036     | 4399    | 0.4958        | 3.0946  |
| satimage-2     | 5803   | 36         | 0.0122  | 0.0000  | 71       | 5732    | 0.6374        | 3.1626  |
| shuttle        | 49097  | 9          | 0.0715  | 0.0000  | 3511     | 45586   | 0.1787        | 56.1602 |
| smtp           | 95156  | 3          | 0.0003  | 0.0000  | 30       | 95126   | 0.2650        | 3.4001  |
| speech         | 3686   | 400        | 0.0165  | 0.0000  | 61       | 3625    | 0.0270        | 0.5902  |
| thyroid        | 3772   | 6          | 0.0247  | 0.0000  | 93       | 3679    | 0.2624        | 3.0971  |
| vertebral      | 240    | 6          | 0.1250  | 0.0000  | 30       | 210     | 0.3866        | 6.7247  |
| vowels         | 1456   | 12         | 0.0343  | 0.0000  | 50       | 1406    | 0.2742        | 1.7779  |
| wbc            | 223    | 9          | 0.0448  | 0.0000  | 10       | 213     | 0.4194        | 3.9709  |
| wine           | 129    | 13         | 0.0775  | 0.0000  | 10       | 119     | 0.2649        | 5.1625  |
| wut/smile      | 1000   | 2          | 0.0000  | 0.0000  | 0        | 1000    | 0.0062        | 2.6009  |
| wut/circles    | 4000   | 2          | 0.0000  | 0.0000  | 0        | 4000    | 0.0061        | 2.1192  |
| wut/isolation  | 9000   | 2          | 0.0000  | 0.0000  | 0        | 9000    | 0.0073        | 1.9025  |
| wut/windows    | 2977   | 2          | 0.0000  | 0.0000  | 0        | 2977    | 0.0273        | 2.1085  |
| wut/x2         | 120    | 2          | 0.0833  | 0.0000  | 10       | 110     | 0.5040        | 3.0701  |
| sipu/compound  | 399    | 2          | 0.0000  | 0.0000  | 0        | 399     | 0.3183        | 2.3987  |
| sipu/jain      | 373    | 2          | 0.0000  | 0.0000  | 0        | 373     | 0.4418        | 2.8734  |
| sipu/flame     | 240    | 2          | 0.0500  | 0.0000  | 12       | 228     | 0.0159        | 2.1107  |
| sipu/spiral    | 312    | 2          | 0.0000  | 0.0000  | 0        | 312     | 0.0012        | 2.2402  |
| fcps/lsun      | 400    | 2          | 0.0000  | 0.0000  | 0        | 400     | 0.0984        | 2.3821  |
| graves/fuzzzyx | 1000   | 2          | 0.0000  | 0.0000  | 0        | 1000    | 0.0072        | 2.9222  |